



Reversible and Irreversible Process

A thermodynamical process taking a system from initial state i to final state f is reversible, if the process can be turned back such that both, the system and the surroundings return to their original states, with no other change anywhere else in the universe.

For a process to be reversible, the following conditions must be satisfied :

1. The process should proceed at an extremely slow rate, i.e., process is quasi-static so that the system is in equilibrium with surroundings at every stage, i.e.
 - (i) The system remains in mechanical equilibrium, i.e., there is no unbalanced force,
 - (ii) The system remains in thermal equilibrium, i.e., all parts of the system and the surroundings remain at the same temperature.
 - (iii) The system remains in chemical equilibrium, i.e., the internal structure of the system does not change.
2. The system should be free from dissipative forces like friction, inelasticity, viscosity, etc. This is because energy spent against such forces cannot be recovered.

As all the conditions mentioned above are of an idealized nature, no process in nature is truly reversible.

Infact, reversibility is an idealized concept which can never be attained. It can at best be approximated.

Some of the examples of approximately reversible processes are :

- (i) An ideal gas allowed to expand slowly and then compressed slowly in a cylinder fitted with frictionless movable piston.
- (ii) Electrolysis can be taken as a reversible process provided resistance offered by electrolyte is zero.
- (iii) Slow compression and expansion of a spring can also be treated as a reversible process.

Now answer the following questions :

Q.1 What are irreversible process

Ans. A process, which does not satisfy any of the conditions for reversible process is called an irreversible process.

Infact, all spontaneous processes of nature are irreversible processes. For example, transfer of heat from a hot body to a cold body, ordinary expansion of a gas, diffusion of gases, stopping of moving body through friction etc. are all irreversible processes.

Q.2 What are fundamental cause of irreversibility ?

Ans. Irreversibility arises mainly from two causes :

- (i) Many processes like free expansion or an explosive chemical reaction take the system to non equilibrium states.
- (ii) Most processes involve friction, viscosity and other dissipative effects.

As the dissipative effects are present everywhere, and they can be minimised only and cannot be fully eliminated, therefore, most processes we deal with are irreversible processes.

Q.3 Give some example of irreversible process.

Ans. Examples of irreversible processes are :

- (i) Most of the chemical reactions are irreversible, because they involve changes in the internal structure of the constituents.
- (ii) The decay of organic matter is an irreversible process.
- (iii) Rusting of iron is an irreversible process.
- (iv) Adiabatic (sudden) compression and expansion of a gas are irreversible processes.

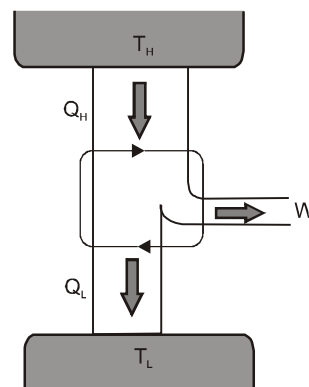




HEAT ENGINES

We have seen that when mechanical work is done on a system, its internal energy increases (remember, we assume that the system does not have any systematic motion). The reverse process in which mechanical work is obtained at the expense of internal energy is also possible. Heat engines are devices to perform this task. The basic activity of a heat engine is shown in figure. It takes some heat from bodies at higher temperature, converts a part of it into the mechanical work and delivers the rest to bodies at lower temperature.

The substance inside the engine comes back to the original state. A process in which the final state of a system is the same as its initial state, is called a cyclic process. An engine works in cyclic process.



Efficiency

Suppose an engine takes an amount Q_H of heat from high-temperature bodies, converts a part W of it into work and rejects an amount Q_L of heat to low-temperature bodies. If the final state of the substance inside the engine is the same as the initial state, there is no change in its internal energy. By first law of thermodynamics, $W = Q_H - Q_L$.

The efficiency of the engine is defined as $\eta = \frac{\text{work done by the engine}}{\text{heat supplied to it}}$

$$= \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

We now describe Carnot engine

The significance of the concept of reversibility :

The main concern of thermodynamics is the efficiency with which heat can be converted into mechanical work. It turned out that a heat engine based on idealised reversible processes achieves the highest possible efficiency. All other engines involving irreversibility of some kind have much lower efficiency because there is energy loss in friction viscous forces etc. Therefore, reversibility is an important concept in thermodynamics.

Comprehension :

A steam engine delivers 5.4×10^8 J of work per minute and takes 3.6×10^9 J of heat per minute from its boiler.

Now answer the following questions :

- Which engine has more efficiency ?
 (A) Reversible engine (B) Irreversible engine
 (C) Both have same efficiency (D) Can't say

Ans. (A)

- What is the efficiency of the engine ?
 (A) 10% (B) 15% (C) 20% (D) 25%

Ans. (B)

- How much heat is wasted per minute ?
 (A) 3.06×10^9 (B) 3.06×10^8 (C) 3.6×10^9 (D) 3.6×10^8

Ans. (A)

- If this heat engine is used as a water pump, then how much water per minute can be transferred at a building of height 30 meter ?
 (A) 1.8×10^6 Kg (B) 1.8×10^5 Kg (C) 5.4×10^4 Kg (D) 5.4×10^3 Kg

Ans. (A)



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Sol. Work done per minute output = 5.4×10^8 J
Heat absorbed per minute, input = 3.6×10^9 J

$$\text{Efficiency, } \eta = \frac{5.4 \times 10^8}{3.6 \times 10^9} = 0.15$$

$$\% \eta = 0.15 \times 100 = 15$$

Heat energy wasted/minute

= Heat energy absorbed/minute – Useful work done/minute

$$= 3.6 \times 10^9 - 5.4 \times 10^8 = (3.6 \times 0.54) \times 10^9 = 3.06 \times 10^9 \text{ J.}$$

$mgh = W$

$$m \times 10 \times 30 = 5.4 \times 10^8$$

$$\Rightarrow m = 1.8 \times 10^6 \text{ kg}$$

Comprehension

The first law of thermodynamics establishes the essential equivalence between the heat energy and mechanical work and says that the two can be converted into each other. Further, 4.18 joule of mechanical work are required to produce one calorie of heat and vice-versa. However, this law has the following limitations :

1. The first law does not indicate the direction in which the change can occur.
For example (i) when two bodies at different temperatures are put in thermal contact with each other, heat flows from the body at higher temperature to the body at lower temperature. We now know that heat cannot flow from the body at lower temperature to the body at higher temperature, although first law of thermodynamics is not violated.
(ii) When a moving car is stopped by applying brakes, work done against friction is converted into heat. When the car cools down, it does not start moving with the conversion of all its heat energy into mechanical work.
(iii) When a bullet strikes a target, kinetic energy of the bullet is converted into heat energy. But heat energy developed in the target cannot be converted back into mechanical energy of the bullet enabling it to fly back.
2. The first law gives no ideal about the extent of change
Our observations and experience tell that there appears to be no restriction on conversion of mechanical work into heat. But there are severe restrictions on the reverse process, i.e., conversion of heat energy into mechanical energy.
We know that heat is not converted into mechanical energy all by itself. An external agency called heat engine is required for the purpose.
No heat engine can convert all the heat energy received from the source into mechanical energy. The first law of thermodynamics is silent about all this.
3. The first law of thermodynamics gives no information about the source of heat, i.e., whether it is a hot or a cold body.
These limitations lead to the formulation of another law called the "**second law of thermodynamics**".

SECOND LAW OF THERMODYNAMICS

This law specifies the conditions for the conversion of heat into work. There are several statements of this law but the following two are the most significant :

(i) Kelvin-Planck Statement

"It is impossible to construct an engine, operating in a cycle, which will produce no effect other than extracting heat from a reservoir and performing an equivalent amount of work ." In simple words, it is not possible to get a continuous supply of work from a body by cooling it to temperature lower than that of the surroundings.

This form of the law is applicable to heat engines. The working substance of a heat engine, operating in a cycle, cannot convert all the extracted heat into work. It must reject a part of the heat to the sink at a lower temperature. So, in order to convert heat into work, it is necessary to have both source and sink. Since all the heat extracted from the source can never be converted into work therefore the efficiency of the engine is never one.



(ii) Rudlope Classius Statement

"It is impossible to make heat flow from a body at a lower temperature to a body at a higher temperature without doing external work on the working substance". In simple words, heat cannot by itself, flow from a body at a lower temperature to a body at a higher temperature.

This form of the law is applicable to ice plants and refrigerators. The refrigerant absorbs heat from inside the refrigerator and rejects a greater quantity of heat to the surroundings (at higher temperature) with the help of an external agency say an electric motor. In ammonia ice plant heat is absorbed from the brine solution at a lower temperature and rejected into water at a higher temperature. This is achieved with the help of an external agency like a pump.

Now answer the following questions :

1. What forbids the complete conversion of work into heat?

Ans. Second law of thermodynamics.

2. Can mechanical work be completely converted into heat. Is reverse also possible?

Ans. The mechanical work can be completely converted into heat but heat extracted from some body cannot be completely converted into useful work.

3. **Statement-1 :** It is not possible for a system unaided by any external agency to transfer heat from a body at lower temperature to another body at higher temperature.

Statement-2 : It is not possible to violate the second law of thermodynamics.

Then which of the combination is true.

- (A) T, T (B) T, F (C) F, T (D) F, F

Ans. (A)

4. "Heat cannot be itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of :

- (A) second law of thermodynamics (B) conservation of momentum
(C) conservation of mass (D) first law of thermodynamics

Ans. (A)

Sol. Heat cannot flow itself from a lower temperature to a body of higher temperature. This corresponds to second law of thermodynamics.

Comprehension :

Changes in energy within a closed system do not set the direction of irreversible processes. Rather that direction is set by another that we shall discuss here that is change in entropy ΔS of the system.

ENTROPY

Like pressure, volume temperature internal energy etc. we have another thermodynamic variable of a system named entropy. In a given equilibrium state, the system has a definite value of entropy. If the system has a temperature T (in absolute scale) and a small amount of heat ΔQ is given to it, we define the change in the entropy of the system as

$$\Delta S = \frac{\Delta Q}{T} \quad \dots\dots\dots(i)$$

In general, the temperature of the system may change during a process. If the process is reversible, the change in entropy is defined as

$$S_f - S_i = \int_i^f \frac{\Delta Q}{T} \quad \dots\dots\dots(ii)$$

In an adiabatic reversible process, no heat is given to the system, The entropy of the system remains constant in such a process.

Entropy is related to the disorder in the system. Thus, if all the molecules in a given sample of a gas are made to move in the same direction with the same velocity, the entropy will be smaller than that in the actual situation in which the molecules move randomly in all directions.



An interesting fact about entropy is that it is not a conserved quantity. More interesting is the fact that entropy can be created but cannot be destroyed. Once some entropy is created in a process, the universe has to carry the burden of that entropy for ever. The second law of thermodynamics may be stated in terms of entropy as follows.

It is not possible to have a process in which the entropy of an isolated system is decreased.

Now answer the following questions :

1. Which has more entropy a crowd or a military force ?

Ans. Crowd has more entropy due to randomness.

2. What is the change in entropy for an adiabatic process ?

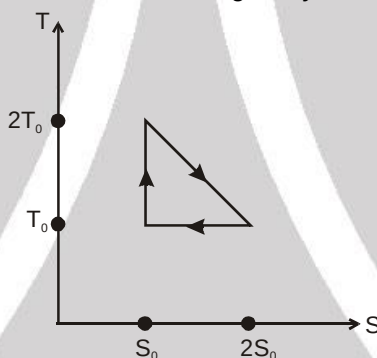
Ans. Zero.

3. When you make ice cubes from water, the entropy of water

- (A) does not change
(B) increases
(C) decreases
(D) may either increase or decrease depending on the process used.

Ans. (C)

4. The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is :



(A) $\frac{1}{2}$

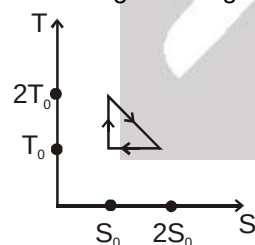
(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

Ans. (C)

Sol. According to the figure



$$Q_1 = T_0 S_0 + \frac{1}{2} T_0 S_0 = \frac{3}{2} T_0 S_0$$

$$Q_2 = T_0 (2S_0 - S_0) = T_0 S_0$$

$$Q_3 = 0$$

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1} = 1 - \frac{2}{3} = \frac{1}{3}$$





Comprehension

We cannot move a ship in an ocean by utilising the energy of the ocean. Why? Explanation is here.

CARNOT'S IDEAL HEAT ENGINE

It is an ideal heat engine which is free from all the imperfections of an actual engine. So, it cannot be realised in actual practice. It was conceived by Niolas Le'onard Sadi Carnot, a French Engineer. This engine serves us a standard by which the performance of actual engines can be judged. It consists essentially of the following parts.

- (i) **Source.** It serves as source of heat. It is maintained at a constant high temperature T_1 K. It has infinite thermal capacity i.e., any amount of heat may be extracted from it at a constant temperature T_1 .
- (ii) **Sink.** It is a cold body maintained at constant low temperature T_2 K. It also has infinite thermal capacity, i.e., any amount of heat rejected to it will not affect its temperature.

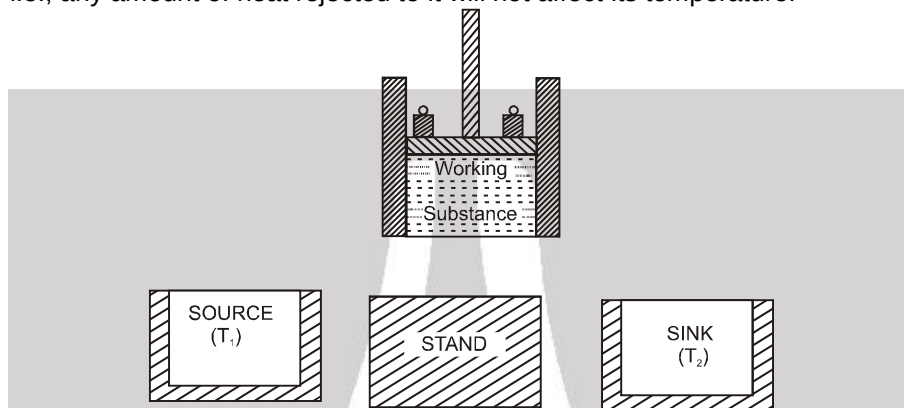


Fig. Carnot's ideal heat engine

(iii) **Insulating stand.** It is a perfectly non-conducting pad.

(iv) **A cylinder :** With perfectly non-conducting walls but with a perfectly conducting bottom. It is fitted with a perfectly non-conducting and frictionless piston over which some weights are placed. One mole of an ideal gas is enclosed in the cylinder. The ideal gas acts as the working substance.

The working substance is subjected to the following four successive reversible operations so as to complete a reversible cycle. This cycle is called Carnot's cycle. The reversibility of operations is a very important assumption because our aim is to find out the maximum efficiency attainable by engine where all sources of irreversibilities are absent.

To begin with let the pressure, volume and temperature be P_1 , V_1 and T_1 respectively. The state of the working substance is represented by the point a in the P-V diagram.

(1) Operation I (Isothermal Expansion). The cylinder is placed on the source. The piston is allowed to move out infinitely slowly by reducing very gradually the weights on the piston. The gas expands extremely slowly. As the gas expands, its temperature tends to fall. But since it is in thermal contact with the heat source therefore it will extract a certain amount of heat Q_1 from the source. In this way the temperature of the gas will remain T_1 throughout the process of expansion. In other words, the gas expands isothermally at temperature T_1 K. This isothermal expansion is represented by the curve AB on the indicator diagram. Let W_1 be the work done by the gas in expanding from volume V_1 to volume V_2 . The pressure decreases from P_1 to P_2 .

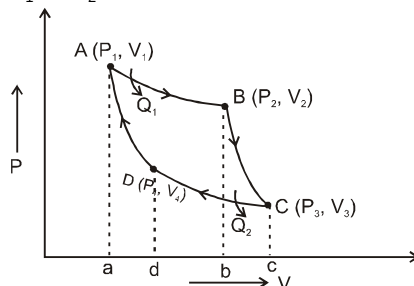


Fig. P-V Diagram of Carnot's cycle

Applying first law of thermodynamics,

$$Q_1 = W_1 = \int_{V_1}^{V_2} P dV = RT_1 \log_e \frac{V_2}{V_1} = \text{Area AB ba A}$$



(2) Operation II (Adiabatic Expansion). The cylinder is placed on the insulated stand and the piston is allowed to move out. The gas expands adiabatically from volume V_2 to volume V_3 till its temperature falls to T_2 K.

The pressure falls from P_2 to P_3 .

The adiabatic expansion is represented by the curve BC in the indicator diagram.

The work done by the gas is given by

$$W_2 = \int_{V_2}^{V_3} PdV = \frac{R}{\gamma - 1} (T_1 - T_2) = \text{Area BC cb B}$$

(3) Operation III (Isothermal Compression) The cylinder is placed on the sink and the gas is isothermally compressed until the pressure and volume become P_4 and V_4 respectively. The operation is represented by the isothermal curve CD. The heat Q_2 developed in compression is absorbed by the sink. Let W_3 be the work done on the gas.

Applying first law of thermodynamics.

$$Q_2 = W_3 = - \int_{V_3}^{V_4} PdV = -RT_2 \log_e \frac{V_4}{V_3} = RT_2 \log_e \frac{V_3}{V_4} = \text{Area C cd DC}$$

(4) Operation IV (Adiabatic Compression) The cylinder is placed on the insulating stand and the gas is compressed adiabatically till it attains its initial pressure P_1 volume V_1 and temperature T_1 . The adiabatic compression is represented by the curve DA in the indicator diagram. Let W_4 be the work done on the gas.

$$\text{Then } W_4 = - \int_{V_4}^{V_1} PdV = \frac{R}{\gamma - 1} (T_1 - T_2) = \text{Area AD da A}$$

Let W be the net external work done by the working substance during one cycle.

Then, $W = \text{Work done by the gas} - \text{Work done on the gas}$

$$= W_1 + W_2 - W_3 - W_4 = W_1 - W_3 \quad [\because W_2 = W_4]$$

$$= \text{area AB ba A} - \text{area C cd DC} = \text{area ABCDA}$$

The working substance can be taken through the cycle again and again. In this way, more and more work can be done by the engine.

Thermal Efficiency of a Carnot engine is defined as the ratio of the external work done in one cycle to corresponding amount of heat extracted from the source.

Since the working substance is restored to its initial state therefore there is no change in its internal energy.

Applying first law of thermodynamics,

$$W = Q_1 - Q_2$$

W , Q_1 and Q_2 are all measured in the same units, i.e., either in units of heat or in units of work.

$$\text{Thermal efficiency, } \eta = \frac{\text{external work done}}{\text{heat extracted}}$$

$$\text{or } \eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\text{Here, } Q_1 = W_1 = RT_1 \log_e \frac{V_2}{V_1} \text{ and } Q_2 = W_3 = RT_2 \log_e \frac{V_3}{V_4}$$

$$\text{Now, } \frac{Q_2}{Q_1} = \frac{RT_2 \log_e \left(\frac{V_3}{V_4} \right)}{RT_1 \log_e \left(\frac{V_2}{V_1} \right)} \quad \text{or} \quad \frac{Q_2}{Q_1} = \frac{T_2 \log_e \left(\frac{V_3}{V_4} \right)}{T_1 \log_e \left(\frac{V_2}{V_1} \right)} \quad \dots\dots\dots(i)$$

The points B and C lie on the same adiabatic.

$$\therefore T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \quad \dots\dots\dots(ii)$$

The points A and D lie on the same adiabatic.

$$\therefore T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1} \quad \dots\dots\dots(iii)$$





Dividing (2) by (3) we get

$$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} \quad \text{or} \quad \frac{V_2}{V_1} = \frac{V_3}{V_4} \quad \text{Put in eq. (i)}$$

Then from equation (1), $\frac{Q_2}{Q_1} = \frac{T_2}{T_1} \Rightarrow \eta = 1 - \frac{T_2}{T_1}$

Results (i) The efficiency of the Carnot's ideal engine is independent of the nature of the working substance. It depends only upon the temperatures of the source and sink. The greater the difference between the two temperatures, higher is the efficiency of the Carnot engine.

(ii) Efficiency is the same for all reversible engines working between temperatures T_1 and T_2 .

(iii) η is always less than one. The value of η can be one only if $T_2 = 0$ i.e., if the sink is at absolute zero of temperature. Since the absolute zero of temperature cannot be attained therefore η cannot be equal to one.

(iv) When $T_2 = T_1$, then $\eta = 0$.

So, heat cannot be converted into work without a temperature difference. In other words, heat can be converted into work only if a sink at a lower temperature is available. This explains as to why the large amount of heat energy of sea water cannot be used for deriving mechanical work.

SOLVED EXAMPLE

Example 1. How is the efficiency of a Carnot engine affected by the nature of the working substance?

Solution : The efficiency is independent of the nature of the working substance.

Example 2. A Carnot engine operates between 227°C and 127°C . If it absorbs 60×10^4 calorie at higher temperature, how much work per cycle can the engine perform.

Solution : $T_1 = (227 + 273) \text{ K} = 500 \text{ K}$

$T_2 = (127 + 273) \text{ K} = 400 \text{ K}$

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{400}{500} = \frac{100}{500} = \frac{1}{5}$$

$$\text{But } \eta = \frac{W}{Q_1} \quad \text{or} \quad W = \eta Q_1$$

$$\begin{aligned} \text{or } W &= (1/5) \times 60 \times 10^4 \text{ cal } [\because Q_1 = 60 \times 10^4 \text{ cal}] \\ &= 12 \times 10^4 \text{ cal} = 12 \times 10^4 \times 4.2 \text{ J } [\because 4.2 \text{ J} = 1 \text{ cal}] = 5.04 \times 10^5 \text{ J} \end{aligned}$$

Example 3. A Carnot cycle is performed by air initially at 927°C . Each stage represents a compression or expansion in the ratio 1 : 32. Calculate (i) the lowest temperature (ii) efficiency of the cycle. Given : $\gamma = 1.4$.

Solution : $T_1 = (927 + 273) \text{ K} = 1200 \text{ K}$

$$\frac{V_1}{V_2} = \frac{1}{6}, \gamma = 1.4$$

$$(i) T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} \quad \text{or} \quad T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$\text{or } T_2 = 1200 \left(\frac{1}{32}\right)^{1.4-1} = 300 \text{ K}$$

$$(ii) \text{ Efficiency, } \eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{1200} = 0.75$$

$$\% \text{ age } \eta = 0.75 \times 100 = 75\%$$



EXERCISE

- Can the Carnot engine be realised in actual practice
- A Carnot's reversible heat engine works between 300K and 600K. In each cycle of operations, the engine draws 1000 J of energy from the source at 600 K. Calculate
 - the energy rejected to the sink at 300 K.
 - the external work done by the engine.
 - the efficiency of the engine.
- A power station uses superheated steam (at high pressure) at approximately 473°C. The cold sink corresponds to the temperature at which steam condenses at atmospheric pressure, that is 100°C. What is the maximum theoretical efficiency for the power station ?
 temperature at which steam condenses at atmospheric pressure, that is 100°C. What is the maximum theoretical efficiency for the power station ?
- A Carnot engine works between ice point and steam point. It is desired to increase the efficiency by 20% (a) making temperature of the source constant (b) making temperature of the sink constant. Calculate the change in temperature in two cases. Which one of these will you prefer and why ?
- Two Carnot engines A and B are operated in series. The first one A receives heat at 800 K and rejects to a reservoir at temperature T K. The second engine B receives the heat rejected by the first engine and in turn temperature T K for the following situations.
 - The outputs of the two engines are equal.
 - The efficiencies of the two engines are equal.
- Draw temperature entropy diagram for Carnot cycle and calculate efficiency & prove it only depends on temperature of hot and cold bodies.
- A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to 32 V, the efficiency of the engine is :
 (A) 0.5 (B) 0.75 (C) 0.99 (D) 0.25
- A Carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by 62 K, its efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively :
 (A) 372 K and 310 K (B) 372 K and 330 K (C) 330 K and 268 K (D) 310 K and 248 K
- Which statement is incorrect ?
 (A) All reversible cycles have same efficiency
 (B) Reversible cycle has more efficiency than an irreversible one
 (C) Carnot cycle is a reversible one
 (D) Carnot cycle has the maximum efficiency in all cycles
- Even Carnot engine cannot give 100% efficiency because we cannot :
 (A) prevent radiation (B) find ideal sources
 (C) reach absolute zero temperature (D) eliminate friction
- A Carnot engine takes 3×10^6 cal of heat from a reservoir at 627°C and gives it to a sink at 27°C. The work done by the engine is :
 (A) 4.2×10^6 J (B) 8.4×10^6 J (C) 16.8×10^6 J (D) zero
- A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of 500K. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :
 (A) efficiency of Carnot engine cannot be made larger than 50%
 (B) 1200 K
 (C) 750 K
 (D) 600 K



Assertion and Reason Problems

In each problem a statement of assertion (A) is given and a corresponding statement of reason (R) is given just below it. Of the statements mark the correct answer as :

- (A) If both A and R are true and R is the correct explanation of A.
- (B) If both A and R are true but R is not the correct explanation of A.
- (C) If A is true but R is false.
- (D) If both A and R are false.
- (E) If A is false but R is true.

13. **Assertion (A) :** The efficiency of a carnot engine is determined mainly by the temperature of the source and not by the temperature of the sink

Reason (R) : The efficiency of a carnot engine can be 100% if the temperature of the source is infinite or the temperature of the sink is 0 K.

14. **Assertion (A) :** No heat engine working between two given temperature of source and sink can be more efficient than a perfectly reversible engine working between the same two temperatures.

Reason (R) : Irreversible engines are highly efficient engines.

Comprehension

REFRIGERATOR (HEAT PUMP)

An ideal refrigerator may be regarded as a carnot's ideal heat engine working in the reverse order.

In an actual refrigerator, the vapours of some low boiling point liquid (ammonia or freon – 12) act as the working substance. The working substance absorbs a certain quantity of heat Q_2 from the cold body or sink at lower temperature T_2 . In a household refrigerator, the ice cubes in the freezer compartment and food constitute the cold body. A certain amount of work W is performed by the compressor of the refrigerator on the working substance. The compressor is operated by an electric motor. The quantity of heat Q_1 is rejected to the hot body (atmosphere) at temperature T_1 K by the radiator (fixed at the back of the refrigerator).

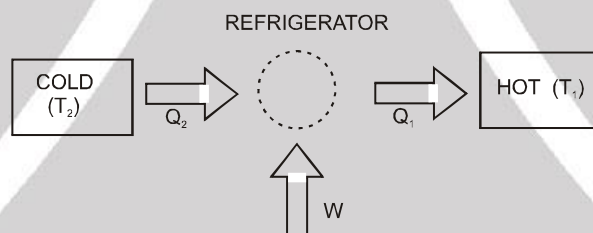


Fig. Refrigerator

Coefficient of performance. It measures the efficiency of a refrigerator.

It is defined as the ratio of the quantity of heat extracted per cycle from the contents of the refrigerator to the mechanical work W done by the external agency to do so.

It is denoted by β or K or ω .

$$\therefore \beta = \frac{Q_2}{W}$$

Smaller the amount of mechanical work done in removing heat Q_2 , greater will be the coefficient of performance.

$$W = Q_1 - Q_2 \quad \therefore \beta = \frac{Q_2}{Q_1 - Q_2} \quad \dots\dots\dots(i)$$

This expression may be put in another form also.

$$\beta = \frac{1}{\frac{Q_1}{Q_2} - 1}$$

[Dividing the numerator and denominator of equation (1) by Q_2 .]



But $\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \therefore \beta = \frac{1}{\frac{T_1}{T_2} - 1} \dots\dots(ii)$

or $\beta = \frac{T_2}{T_1 - T_2}$

Discussion (i) In actual practice, β varies from 2 to 6. For an actual refrigerator, the value of β is less than that calculated from equations (1) or (2).

(ii) Lesser the difference in the temperatures of the cooling chamber and the atmosphere, higher is the coefficient of performance of the refrigerator.

(iii) In a heat engine the efficiency can never exceed 100%. But in the case of a refrigerator, the coefficient of performance may be much higher than 100%

(iv) As the refrigerator works, T_2 goes on decreasing due to formation of too much ice. There is practically no change in T_1 . This decreases the value of β . However, if the refrigerator is defrosted, T_2 shall increase and consequently the value of β . So, it is necessary to defrost the refrigerator.

Solved Example :

1. Refrigerator transfers heat from a cold body to a hot body. Does this not violate the second law of thermodynamics

Ans. No. This is because external work is being performed.

2. Is coefficient of performance of a refrigerator constant?

Ans. No, the coefficient of performance of refrigerator decreases with decrease in its inside temperature.

Now answer the following Questions :

1. Can we increase the coefficient of performance of a refrigerator by increasing the amount of working substance

- (A) No (B) Yes
(C) some time yes, some time no (D) can't say

Ans. (A)

2. The door of an operating refrigerator is kept open in a closed room. Then temperature of room will be :

- (A) increases slightly (B) decreases slightly
(C) may increase or decrease (D) can't say

Ans. (A)

Ans. The room will be slightly warmed.

3. A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

- (1) 99 J (2) 90 J (3) 1 J (4) 100 J

Ans. (2)

Sol. For Carnot engine using as refrigerator

$$W = Q_2 \left(\frac{T_1}{T_2} - 1 \right)$$

It is given $\eta = \frac{1}{10} \Rightarrow \eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = \frac{9}{10}$

So, $Q_2 = 90 \text{ J}$ (as $W = 10 \text{ J}$)



EXERCISE

15. Calculate the coefficient of performance of a refrigerator working between -3°C and 27°C .
16. A refrigerator is to maintain eatables kept inside at 9°C . If room temperature is 36°C , calculate the coefficient of performance.
17. A carnot refrigerator works between temperature limits of 0°C and 27°C . If 10 kg of water at 0°C is converted into ice at 0°C , calculate (a) the heat rejected (b) the energy supplied to the refrigerator. Given : latent heat of ice = 80 kcal kg^{-1} .

1. **It does not indicate the direction in which the change can proceed.**

Illustrations.

(i) When a hot body is brought in thermal contact with a cold body, heat always flows from the hot body to the cold body. Why heat does not flow from the cold body to the hot body? The first law of thermodynamics is silent about it. So, this law does not indicate the direction of heat transfer.

(ii) It is not possible for a ship to use the huge amounts of heat of the sea waters to operate its engine. What prevents the conversion of heat into work? The first law is again silent about this. So, the first law of thermodynamics does not specify the conditions under which heat is converted into work.

2. **The first law of thermodynamics gives no idea about the extent to which the change takes place.**

It has been observed that no heat engine can convert all the heat extracted from the source into mechanical energy. Why the whole of the heat cannot be converted into mechanical energy. The first law is silent about this question.

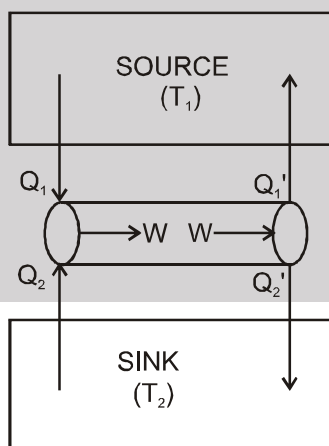
CARNOT'S THEOREM

Carnot showed that no engine can be more efficient than the perfectly reversible engine. This is known as Carnot's theorem and is stated as follows :

No heat engine working between given temperatures can have efficiency greater than that of a reversible engine working between the same temperatures.

Consider two engines – an irreversible engine A and a reversible engine B. Let the two engines be coupled such that A drives B backwards. So B is acting as a refrigerator.

A absorbs heat Q_1 from the source performs work W and rejects heat Q_2 to the sink.



$$\text{Efficiency of A, } \eta_A = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

B absorbs heat Q_2 , from the sink gets work W performed on it and rejects heat Q_1 to the source.

$$\text{Efficiency of B, } \eta_B = \frac{W}{Q_1'} = \frac{Q_1' - Q_2'}{Q_1'}$$

Suppose A is more efficient than B.

Then, $\eta_A > \eta_B$



$$\therefore \frac{W}{Q_1} > \frac{W}{Q_1'} \quad \text{or} \quad Q_1 < Q_1'$$

$\therefore Q_1 - Q_1'$ is positive. So the net quantity of heat given to the source at temperature T_1 by the compound engine AB is $(Q_1' - Q_1)$.

$$\text{Again, } W = Q_1 - Q_2 = Q_1' - Q_2' \quad \text{or} \quad Q_2' - Q_2 = Q_1' - Q_1$$

$$\therefore Q_2' - Q_2 \text{ is positive.} \quad [\because Q_1' - Q_1 \text{ is +ve}]$$

So, the net quantity of heat taken from the sink at temperature T_2 by the compound engine AB is $(Q_2' - Q_2)$.

Thus, the compound engine AB is a self-acting device which is transferring heat from lower temperature to higher temperature without any work being done by the external agency. This is forbidden by second law of thermodynamics. So, our assumption that engine A is more efficient than B is wrong.

This proves Carnot's theorem.

Example 1. Three ideal engines operate between reservoir temperatures of (a) 400 K and 500 K; (b) 600 K and 800 K, and (c) 400 K and 600 K. Rank the engines according to their thermal efficiencies, greatest first.

Answer : c, b, a.

$$\text{Solution :} \quad (c) \quad \eta = 1 - \frac{400}{600} = 1 - \frac{2}{3} = \frac{1}{3} \quad (b) \quad \eta = 1 - \frac{600}{800} = 1 - \frac{3}{4} = \frac{1}{4} \quad (a) \quad \eta = 1 - \frac{400}{500} = 1 - \frac{4}{5} = \frac{1}{5}$$

Example 2. Five moles of an ideal gas are taken in a Carnot engine working between 100°C and 30°C . The useful work done in one cycle is 420 joule. Calculate the ratio of the volume of the gas at the end and beginning of the isothermal expansion. Give : $R = 8.4 \text{ J mol}^{-1} \text{ K}^{-1}$.

$$\text{Solution :} \quad T_1 = (100 + 273) \text{ K} = 373 \text{ K}$$

$$T_2 = (30 + 273) \text{ K} = 303 \text{ K}$$

$$\text{Useful work, } W = Q_1 - Q_2 = 420 \text{ J} \quad \dots\dots(i)$$

Let W_1 and W_2 be the works done during isothermal expansion at T_1 and isothermal compression at T_2 .

$$\text{Now } \frac{Q_1}{Q_2} = \frac{W_1}{W_2} = \frac{T_1}{T_2} = \frac{373}{303}$$

$$\therefore Q_1 = \frac{373}{303} Q_2$$

$$\text{From equation } W = \frac{373}{303} Q_2 - Q_2$$

$$\text{or } 420 = \left(\frac{373}{303} - 1 \right) Q_2 \quad \text{or} \quad Q_2 = 1818 \text{ J}$$

$$\text{Again : } Q_1 = \frac{373}{303} Q_2 = \frac{373}{303} \times 1818 \text{ J} = 2238 \text{ J},$$

Heat used in isothermal expansion is given by

$$Q_1 = \mu RT \log_e \frac{V_2}{V_1} = 2.3026 \times \mu RT \log_{10} \frac{V_2}{V_1}$$

$$\text{or } 2238 = 2.3026 \times 5 \times 8.4 \times 373 \log_{10} \frac{V_2}{V_1}$$

$$\text{On simplification, } \frac{V_2}{V_1} = 1.15$$



EXERCISE

18. What type of process is carnot's cycle
19. One mole of an ideal gas is taken in a carnot engine working between 27°C and 227°C . The useful work done in one cycle is 600J. Calculate the ratio of volume of gas at the end and beginning of the isothermal expansion. Given : $R = 8.3 \text{ J mole}^{-1} \text{ K}^{-1}$.
20. A carnot engine takes in 1000 k cal of heat from a reservoir at 627°C and exhausts heat to sink at 27°C . What is its efficiency? When will its efficiency be 100% ?

Example 1. You wish to increase the coefficient of performance of an ideal refrigerator. You can be do so by (a) running the cold chamber at a slightly higher temperature, (b) running it at a slightly lower temperature, (c) moving the unit to a slightly warmer room, or (d) moving it to a slightly cooler room. The temperature changes are to be the same in all four cases. List the changes according to the resulting coefficients of performance, greatest first.

Answer : (a), (b), (c), (d)

Solution : Closer the temperatures of the two reservoirs to each other higher is the value of coefficient of performance.

Example 2. Assuming a domestic refrigerator as reversible engine working between melting point of ice and the room temperature of 27°C , calculate the energy in joule that must be supplied to freeze one kg of water. Given : temperature of water – 0°C , $L = 80 \text{ cal g}^{-1}$.

Solution : $T_1 = (27 + 273) \text{ K} = 300 \text{ K}$

$T_2 = (0 + 273) \text{ K} = 273 \text{ K}$

Heat to be removed, $Q_2 = mL$

where m is the mass of water and L is the latent heat.

$\therefore Q_2 = 1000 \times 80 \text{ cal} = 8 \times 10^4 \text{ cal}$

[$\therefore m = 1 \text{ kg} = 1000 \text{ g}$]

$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$ or $Q_1 = \frac{T_1}{T_2} \times Q_2$ or $Q_1 = \frac{300}{273} \times 8 \times 10^4 \text{ cal} = 87912.1 \text{ cal}$

Energy required to be supplied, $W = Q_1 - Q_2$

or $W = (87912.1 - 80,000) \text{ cal} = 7912.1 \text{ cal}$
 $= 7912.1 \times 4.2 \text{ J} = 33230.8 \text{ J}$

EXERCISE

21. How much energy in watt-hour may be required to convert 2 kg of water into ice at 0°C assuming that the refrigerator is ideal? Given : temperature of freezer = -15°C , room temperature = 25°C and initial temperature of water = 25°C .

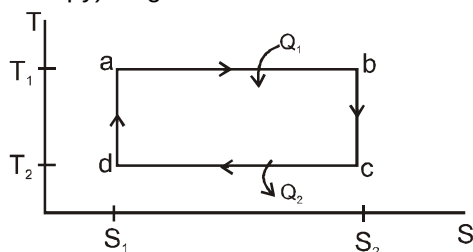
ANSWER KEY OF EXERCISE

1. No. It is an ideal heat engine
2. (i) 500 J (ii) 500 J (iii) 50%
3. 50%
4. (a) 20°C (b) 129.5°C ; Process (a) In which temperature of sink has to be reduced by 20°C will be preferable
5. (i) 550 K (ii) 489.9 K





6. The basic process of a Carnot engine, described above, is again shown in figure in a T-S (temperature-entropy) diagram.



The points a, b, c and d represent the same states as in figure. Let the entropy in state a be S_1 . An amount Q_1 of heat is supplied to the system in the isothermal process ab at the temperature T_1 . The entropy increases in this part as heat is supplied to the system. Also, by definition,

$$S_2 - S_1 = \frac{Q_1}{T_1} \quad \text{.....(i)}$$

The entropy remains constant in the part bc as it describes an adiabatic process. So the entropy in state c is S_2 . In the part cd, the system gives a heat Q_2 at the lower temperature T_2 and its entropy is decreased. The part da represents an adiabatic process and the entropy remains constant. As the entropy in state a is S_1 , the entropy in state d is also S_1 . Using the definition of change in entropy for the process cd,

$$S_1 - S_2 = \frac{Q_2}{T_2} \quad \text{.....(ii)}$$

From (i) and (ii),

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \quad \text{or} \quad \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

The efficiency of the engine is

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} \quad \text{.....(iii)}$$

Thus, the efficiency of the engine depends only on the temperatures of the hot and cold bodies between which the engine works.

7. (B) 8. (A) 9. (A) 10. (C) 11. (B) 12. (C) 13. (E)
14. (C) 15. 9 16. $T_1 = 36^\circ\text{C} = (36 + 273) \text{ K} = 309 \text{ K}$, $T_2 = 9^\circ\text{C} = (9 + 273) \text{ K} = 282 \text{ K}$

$$\text{Coefficient of performance} = \frac{T_2}{T_1 - T_2} = \frac{282}{309 - 282} = \frac{282}{27} = 10.4$$

17. 879.1 kcal, $3.32 \times 10^5 \text{ J}$
(i) Production of heat on passing an electric current through a wire. The same amount of heat is produced in a given time, even when the direction of current is reversed.
(ii) Mixing of two different substances like water and alcohol.
(iii) The formation of solution of a solid in water etc.
18. Cyclic process.
19. 1.43
20. 66.67
The efficiency will be 100 % if $T_2 = 0$, i.e., if the sink is maintained at absolute zero of temperature.
21. 37.98 watt hour